

SMART CONTRACT SECURITY AUDIT - SECUREVOTEMULTISIG

One-Page Visual Audit Report

Purpose: Multi-signature ETH execution contract based on owner approvals.

Risk	Severity	SWC	OWASP
execute() can be replayed because txHash is never marked executed	Critical	SWC-121	SC02 Business Logic
txHash is not bound on-chain to destination/value, enabling approval misuse	High	SWC-105	SC02 Business Logic
Constructor does not validate required threshold or duplicate/zero owners	High	N/A	SC05 Input Validation
approve() is public to non-owners, polluting approval state	Low	N/A	SC01 Access Control

Root Cause: The multisig approval model lacks transaction binding, execution state, and constructor invariant checks.

Business Logic Findings: A valid approval set can authorize repeated or different transfers if txHash governance is misused. Invalid thresholds can permanently break execution.

BENCHMARK METRICS

Time_seconds: 75 | Confidence: 0.95 | Total_findings: 4

Critical: 1 | High: 2 | Medium: 0 | Low: 1

Precision_estimate: 0.93 | Coverage_estimate: 1.00 | Risk_score: 89/100

GOLD STANDARD EVALUATION

TP: N/A | FP: N/A | FN: N/A | Precision: N/A | Recall: N/A | F1: N/A